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Education

University of Washington, Ph.D. Aquatic and Fishery Sciences 2003-2006
- Committee: Ray Hilborn (advisor), Andre Punt, Tim Essington

Montana State University, MSc Biology 2000-2003
- Advisor: Dan Goodman

University of California San Diego, BSc Ecology and Evolution 1995-1999

Employment

Statistician Northwest Fisheries Science Center, NMFS (NOAA) 2009 - present

Affiliate Professor University of Washington (SAFS) 2013 - present

Post-doctoral researcher National Research Council 2007 - 2008

Stock assessment scientist Wellington, New Zealand Spring 2005

Graduate research assistant University of Washington 2003 - 2006

Journal Articles (peer - reviewed)

164. Ford, M.J., **E.J. Ward**, M. Kardos, K.M. Parsons, C. Emmons, and M.B. Hanson. 2026. Perspective: the future of the southern resident killer whales depends on interactions with other killer whale populations. *Ecology and Evolution*, 16:e73205. [link](#)
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161. Liu, O.R., A.O. Shelton, A. Ramón-Laca, K.M. Nichols, **E.J. Ward**, E.M. Phillips, J.E. Zamon, A. Wells, and R.P. Kelly. 2026. Mapping the marine distribution of eulachon (*thaleichthys pacificus*) in the northeast pacific using environmental dna. *Communications Biology*, 9:465. [link](#)
160. Maioli, F., P.D.v. Denderen, M. Lindmark, M. Montanyès, **E.J. Ward**, S.C. Anderson, and M. Lindegren. 2026. Thermal niche warming is more consistent than range shifts in marine species under climate change. *bioRxiv*, :. [link](#)

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158. Indivero, J., S.C. Anderson, L.A.K. Barnett, J.E. Pohl, S.K. Rohan, S. Siedlecki, **E.J. Ward**, and T.E. Essington. 2026. Skill testing oxygen data for distribution modeling of marine species. *Fisheries Oceanography*, 35:168-188. [link](#)
157. **Ward, E.J.**, M.A. Head, K. Ono, K.F. Johnson, and K.N. Marshall. 2026. Applying information theory to increase the efficiency of life-history sampling in fisheries data. *Fisheries Research*, 293:107641. [link](#)
156. Oken, K.L., S.D. Groth, D.S. Holland, A.E. Punt, and **E.J. Ward**. 2025. Variability in somatic growth over time and space determines optimal season-opening date in the oregon ocean shrimp (*pandalus jordani*) fishery. *Canadian Journal of Fisheries and Aquatic Sciences*, 82:1-13. [link](#)
155. Odell, A.N., K.N. Marshall, **E.J. Ward**, K.F. Johnson, and M.L. Baskett. 2025. Spatiotemporal models reveal dynamic growth patterns in u.s. west coast groundfish. *Transactions of the American Fisheries Society*, 155:76-89. [link](#)
154. **Ward, E.J.**, P.A. English, C.N. Rooper, B.E. Ferriss, C.E. Whitmire, C.R. Wetzel, L.A.K. Barnett, S.C. Anderson, J.T. Thorson, K.F. Johnson, J. Indivero, and E.H. Markowitz. 2025. surveyjoin: a standardized database of scientific trawl surveys in the northeast pacific ocean. *PeerJ*, 13:e19964. [link](#)
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151. Indivero, J., S.C. Anderson, L.A.K. Barnett, T.E. Essington, and **E.J. Ward**. 2025. Estimating a physiological threshold to oxygen and temperature from marine monitoring data reveals challenges and opportunities for forecasting distribution shifts. *Ecography*, 2025:eo7413. [link](#)
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144. Samhouri, J.F., B.E. Feist, M. Jacox, O.R. Liu, K. Richerson, E. Steiner, J. Wallace, K. Andrews, L. Barnett, A.H. Beaudreau, L. Bellquist, M.P. Buil, M.A. Haltuch, A. Harley, C.J. Harvey, I.C. Kaplan, K. Norman, A. Phillips, L.K. Rasmuson, **E.J. Ward**, C. Whitmire, and R.L. Selden. 2024. Stay or go? geographic variation in risks due to climate change for fishing fleets that adapt in-place or adapt on-the-move. *PLOS Climate*, 3:1-28. [link](#)
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134. Davies, S.C., P.L. Thompson, C. Gomez, J. Nephin, A. Knudby, A.E. Park, S.K. Friesen, L.J. Pollock, E.M. Rubidge, S.C. Anderson, J.C. Iacarella, D.A. Lyons, A. MacDonald, A. McMillan, **E.J. Ward**, A.M. Holdsworth, N. Swart, J. Price, and K.L. Hunter. 2023. Addressing uncertainty when projecting marine species distributions under climate change. *Ecography*, :e06731. [link](#)
133. Jensen, A.J., R.P. Kelly, W.H. Satterthwaite, **E.J. Ward**, P. Moran, and A.O. Shelton. 2023. Modeling ocean distributions and abundances of natural- and hatchery-origin chinook salmon stocks with integrated genetic and tagging data. *PeerJ*, 11:e16487. [link](#)
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- vary among pacific salmon species and populations. *Nature Ecology & Evolution*, 7:852–861. [link](#)
126. Buckner, J.H., W.H. Satterthwaite, B.W. Nelson, and **E.J. Ward**. 2023. Interactions between life history and the environment on changing growth rates of chinook salmon. *Canadian Journal of Fisheries and Aquatic Sciences*, 80:648-662. [link](#)
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 124. May, S.A., J.J. Hard, M.J. Ford, K.A. Naish, and **E.J. Ward**. 2023. Assortative mating for reproductive timing affects population recruitment and resilience in a quantitative genetic model. *Evolutionary Applications*, 16:657-672. [link](#)
 123. Davis, M.J., J. Anthony, **E.J. Ward**, J. Firman, and C. Lorion. 2023. Coherence among Oregon Coast coho salmon populations highlights increasing relative importance of marine conditions for productivity. *Fisheries Oceanography*, 32:293-310. [link](#)
 122. Good, T.P., J.E. Jannot, K.A. Somers, and **E.J. Ward**. 2022. Using Bayesian time series models to estimate bycatch of an endangered albatross. *Fisheries Research*, 256:106492. [link](#)
 121. **Ward, E.J.**, L.A. Barnett, S.C. Anderson, C.J. Commander, and T.E. Essington. 2022. Incorporating non-stationary spatial variability into dynamic species distribution models. *ICES Journal of Marine Science*, 79:2422–2429. [link](#)
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 119. Brodie, S., J.A. Smith, B.A. Muhling, L.A.K. Barnett, G. Carroll, P. Fiedler, S.J. Bograd, E.L. Hazen, M.G. Jacox, K.S. Andrews, C.L. Barnes, L.G. Crozier, J. Fiechter, A. Fredston, M.A. Haltuch, C.J. Harvey, E. Holmes, M.A. Karp, O.R. Liu, M.J. Malick, M.P. Buil, K. Richerson, C.N. Rooper, J. Samhour, R. Seary, R.L. Selden, A.R. Thompson, D. Tommasi, **E.J. Ward**, and I.C. Kaplan. 2022. Recommendations for quantifying and reducing uncertainty in climate projections of species distributions. *Global Change Biology*, 28:6586-6601. [link](#)
 118. Ettinger, A., C. Harvey, J. Samhour, C. Emmons, M. Hanson, **E.J. Ward**, and J. Olson. 2022. Shifting phenology of an endangered apex predator tracks changes in its favored prey. *Endangered Species Research*, 48:211-223. [link](#)
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111. Hunsicker, M., **E.J. Ward**, M. Litzow, S. Anderson, C. Harvey, J. Field, J. Gao, M. Jacox, S. Melin, A. Thompson, and P. Warzybok. 2022. Tracking and forecasting community responses to climate perturbations in the california current ecosystem. *PLOS Climate*, 1:e0000014. [link](#)
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105. Waples, R.S., R.K. Waples, and **E.J. Ward**. 2021. Pseudoreplication in genomics-scale datasets. *Molecular Ecology Resources*, 22:503–518. [link](#)
104. Stewart, J.D., J.W. Durban, H. Fearnbach, L.G. Barrett-Lennard, P.K. Casler, **E.J. Ward**, and D.R. Dapp. 2021. Survival of the fattest: linking body condition to prey availability and survivorship of killer whales. *Ecosphere*, 12:e03660. [link](#)
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- approach to stock size estimation. *PLOS ONE*, 16:e0241254. [link](#)
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 15. Eguchi, T., C. Fahy, J. Jannot, K. Somers, and **E. Ward**. 2017. 2016 Leatherback sea turtle estimated bycatch reporting requirements as set out in the NMFS Biological Opinion for the continuing authorization of the Pacific Coast groundfish fisheries. National Marine Fisheries Service, SWFSC, La Jolla, CA 92039. [link](#)
 14. Good, T. P., **E. Ward**, J. Jannot, R. Shama, N. Riley, and J. McVeigh. 2017. Observed and Estimated Bycatch of Short-tailed Albatross in U.S. West Coast Groundfish Fisheries 2014-2015. National Marine Fisheries Service, NWFSC, 2725 Montlake Blvd E., Seattle, WA 98112. (link)[https://www.pcouncil.org/wp-content/uploads/2017/03/F5a_NMFS_Rpt6_ElectricOnly_STAL_bycatch_report_2017_Apr2017BB.pdf]
 13. Gustafson, R., Y.-W. Lee, **E. Ward**, K. Somers, V. Tuttle, J. Jannot, and J. McVeigh. 2017. Observed and Estimated Bycatch of Eulachon in 2002–2015 US West Coast Groundfish Fisheries. National Marine Fisheries Service, NWFSC, 2725 Montlake Blvd E., Seattle, WA 98112. [link](#)
 12. Lee, Y.-W., R. Gustafson, J. Jannot, J. McVeigh, N. Riley, K. Somers, V. Tuttle, S. Wang, and **E. Ward**. 2017. Observed and Estimated Bycatch of Green Sturgeon in 2002–2015 US West Coast Groundfish Fisheries. National Marine Fisheries Service, NWFSC, 2725 Montlake Blvd E., Seattle, WA 98112. [link](#)

11. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2016 Elwha River Steelhead Escapement Estimate Based on DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.
10. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2016 Elwha River Chinook Escapement Estimate Based on DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.
9. Gustafson, R.G., L. Weitkamp, Y.W. Lee, **E.J. Ward**, K. Somers, V. Tuttle, and J. Jannot. 2016. Status Review Update of Eulachon (*Thaleichthys pacificus*) Listed under the Endangered Species Act: Southern Distinct Population Segment. [link](#)
8. Gerrodette, T., **E.J. Ward**, R.L. Taylor, L.K. Schwarz, T. Eguchi, P.R. Wade, and G.K. Himes-Boor. 2016. Daniel Goodman's empirical approach to Bayesian statistics. Pre-print at PeerJ [link](#)
7. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2015 Elwha River Steelhead Escapement Estimate Based on DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.
6. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2015 Elwha River Chinook Escapement Estimate Based on DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.
8. Eguchi, T., R. Bjorkland, C. Fahy, J. Jannot, T. Mitchell, K. Somers, **E. Ward**, B. Whiteside. 2015. Reporting requirements as set out in the NMFS Biological Opinion Leatherback sea turtles for the continuing authorization of the Pacific Coast groundfish fisheries. National Marine Fisheries Service, SWFSC, La Jolla, CA 92039. [link](#)
7. Gustafson, R., Y.-W. Lee, **E. Ward**, K. Somers, V. Tuttle, J. Jannot. 2015. Observed and Estimated Bycatch of Eulachon in 2002–2013 US West Coast Groundfish Fisheries. National Marine Fisheries Service, NWFSC, 2725 Montlake Blvd E., Seattle, WA 98112. [link](#)
6. Lee, Y.-W., R. Gustafson, J. Jannot, J. McVeigh, N. Riley, K. Somers, V. Tuttle, S. Wang, **E. Ward**. 2015. The next link/button will exit from NWFSC web site Observed and Estimated Bycatch of Green Sturgeon in 2002–2013 US West Coast Groundfish Fisheries. National Marine Fisheries Service, NWFSC, 2725 Montlake Blvd E., Seattle, WA 98112. [link](#)
5. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2014 Elwha River Steelhead Escapement Estimate Based on DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.
4. Denton, K., M. McHenry, R. Moses, **E.J. Ward**, M. Liermann, O. Stefankiv, W. Wells and G. Pess. 2014 Elwha River Chinook Escapement Estimate Based on

DIDSON/ARIS Multi-Beam SONAR Data. Lower Elwha Tribe, Port Angeles, Washington.

3. **E.J. Ward**, M.J. Ford, R.G. Kope, J.K.B. Ford, A. Velez-Espino, C.K. Parken, L. LaVoy, M.B. Hanson, K.C. Balcomb. Estimating the impacts of Chinook salmon abundance and prey removal by ocean fishing on Southern Resident killer whale population dynamics. 2013. U.S. Dept. Commer., NOAA Tech. Memo. NMFS-NWFSC-123, 71 p. [link](#)
2. Stout, H.A., P.W. Lawson, D.L. Bottom, T.D. Cooney, M.J. Ford, C.E. Jordan, R.G. Kope, L.M. Kruzic, G.R. Pess, G.H. Reeves, M.D. Scheuerell, T.C. Wainwright, R.S. Waples, **E.J. Ward**, L.A. Weitkamp, J.G. Williams, and T.H. Williams. 2012. Scientific conclusions of the status review for Oregon coast coho salmon (*Oncorhynchus kisutch*). U.S. Dept. Commer., NOAA Tech. Memo. NMFS-NWFSC-118, 242 p. [link](#)
1. Northwest Fisheries Science Center. 2011. Risk assessment of U.S. West Coast groundfish fisheries to threatened and endangered marine species. NWFSC, NMFS. 2725 Montlake Blvd E, Seattle, WA. [link](#)

Book Chapters

E.J. Ward 2012. Population Dynamics. In *Grzimek's Animal Life Encyclopedia*. pp. 195-208.

Teaching Experience

FISH 507: Applied Time Series Analysis, University of Washington 2012 - present
Co-taught course with Eli Holmes and Mark Scheuerell [webpage](#)

Ecological Society of America (ESA) 2008 - 2011
Co-taught Analysis of Multivariate Time-Series Data Using State-Space Models, and Introduction to MARSS

NWFSC R workshop (organized & taught) 2010 - 2011

Visiting lecturer University of Washington 2007 - 2009
Taught FSH 497 Introductory modeling and programming course

External Funding

2026 J. Nielsen et al. New phenology and lower trophic indicators for predicting fish recruitment in the Gulf of Alaska, North Pacific Research Board flex grant. \$140K.

2025 K. Oken et al. Ground truthing model-based fishery-dependent indices for West Coast groundfish fisheries. NMFS Cooperative Research Program. \$150K

2024 K. Oken and E.J. Ward. Linking ecosystem indicators to improve fisheries stock assessments. NWFSC RFP, \$225K

2024 E.J. Ward et al. Understanding the impacts of marine heatwaves on primary production in the California Current. NWFSC RFP, \$245K

2023 E.J. Ward, K.N. Marshall. Developing multispecies models to improve forecasts and identify robust ecosystem management strategies under climate change. NMFS international fellowship to collaborate with Norway's Institute of Marine Research. \$36K

2022-2023 E.J. Ward, M. Hunsicker, B. Burke, W. Satterthwaite, M. Jacox, M. Litzow, M. Malick. *Non-stationary relationships between climate and fisheries in the California Current*. NWFSC grant, \$118K

2022-2024 T. Essington, M.E. Hunsicker, S.C. Anderson, E.J. Ward, J.T. Thorson, S. Siedlecki, L. Barnett, R. Forrest. *Climate change, distribution shifts, and choke species: A threat assessment to advance fishery ecosystem planning*. Washington Sea Grant, \$250K

2021 E.J. Ward. *Characterizing range shifts in groundfish species in response to local climate velocities*. NMFS Office of Science and Technology, \$40K.

2021-2023 C.E. Jordan, M.E. Hunsicker, and E.J. Ward. *Developing a dashboard for salmon data and ecosystem indicators*. NOAA Protected Species Toolbox, \$150K

2019-2021 E.J. Ward, A.O. Shelton, R. Sharma, B. Nelson. *A spatially explicit model for evaluating threats to coho salmon in freshwater and marine life stages in the Northeast Pacific Ocean*. Pacific Salmon Commission, \$229K

2019-2021 D. Finnoff, E.J. Ward, T. Miller, T. Schwoerer, M. Horning, D. Dugan. *Thresholds in a changing ocean environment: bioeconomic implications to inform adaptation decisions for Alaska's salmon fisheries*. NOAA Ocean Acidification Program, \$1,050,000.

2018-2020 A.O. Shelton, J. Watson, C. Cunningham, E.J. Ward, A. Grey, M.J. Ford, K. Sommers. *Integrating Chinook Catches from Directed Fisheries and Trawl Bycatch to Improve Distribution Information for Bycatch Avoidance and Impacts On Predators in a Changing Environment*. NOAA Protected Species, \$200K.

2018-2020 E.J. Ward, C. Harvey, J. Samhouri, etc. *Spatial ecosystem state and pressure indicators and scale-dependence of temporal indicator performance*. FATE (Fisheries And The Environment), \$192K.

2018-2020 M. Hunsicker, M. Litzow, E.J. Ward, B. Burke, S. Zador, C. Cunningham, and N. Bond. *Measuring the strength of ocean-atmosphere coupling to predict climate forcing of northeast Pacific ecosystems*. FATE (Fisheries And The Environment), \$198K.

2018-2020 G. Holtgrieve, C. Harvey, and E.J. Ward. *Reconstructing a Century of Coastal Productivity and Predator Trophic Dynamics using Compound-specific Stable Isotopes from Archival Bone Specimens*. Washington Sea Grant, \$192K.

2018-2020 L. Hauser, B. Vadopalas, R. Childer, B. Hudson, B. Eudeline, and E.J. Ward. *Development of genetic risk assessment tools and Management Strategy Evaluation for Aquaculture of Native Shellfish*. Saltonstall-Kennedy Grant, \$299K.

2017-2018 E.J. Ward, G. Holtgrieve, C. Harvey. *Reconstructing a ecosystem productivity and marine mammal trophic dynamics using compound-specific stable isotopes from bone specimens*. NWFSC Internal Grants Program

2016-2018 T.E. Essington, T.B. Francis, C. Greene, L. Kuehne, D. Lowry, E.J. Ward. *Recovering historical baselines in Puget Sound*. Washington Sea Grant, \$243K.

2016 E.J. Ward *Improving bycatch estimation through spatial statistics* \$100K.

2015 E.J. Ward, A.O. Shelton and M.J. Ford. *Portfolio Effects in Historic Chinook Hatchery Practices on Prey Availability for Southern Resident Killer Whales* National Fish and Wildlife Foundation, \$110K.

2014 - 2016 *Applying portfolio effects to the Gulf of Alaska ecosystem: Did multi-scale diversity buffer against the Exxon Valdez oil spill?* NCEAS working group, \$150K

2014-2015 I. Kaplan, E.J. Ward, et al. *A spatially explicit ecosystem model for quantifying marine mammal impacts on Chinook salmon* Pacific Salmon Commission Southern Fund, \$206K.

2013-2014 T. Eguchi, E.J. Ward, and E.E. Holmes. *Developing risk assessment tools for marine sea turtles* \$100K.

2013-2014 Holmes, E.E. and E.J. Ward. *Developing web based computing tools for protected species.* \$200K.

2013 M.B. Hanson, E.J. Ward. et al. *Modeling the distribution of endangered killer whales in Pacific Northwest Navy ranges using acoustic recorder detections and satellite tagged whales*, US Navy.

2013 Shelton, A.O., E.J. Ward, J.T. Thorson, M. Bellman, B. Feist. *Integrating spatial habitat and fisheries effort data to improve abundance estimates of west coast groundfish.* \$56K.

2011 E.J. Ward et al., *Incorporating multiple data sources into a unified framework to improve estimates of adult salmon returns and population productivity.* \$33K.

2009 NSF Comparative Analysis of Marine Ecosystem Organization: New statistical tools for analyzing community dynamics with applications to marine zooplankton, \$347K (141K from NCEAS). Associate Investigator (Principal Investigator - E.E. Holmes)

Fellowships and Awards

2022 NOAA Bronze Medal Award, Pacific Fisheries Management Council working group: "For leadership and problem-solving to develop innovative protections for Southern Resident killer whales and provide important fishing opportunities."

2006 *Cumulative Risk Assessment for Endangered Species* 2 year NRC post-doc awarded \$92K.

2003 *National Marine Fisheries Service / Washington Sea Grant Fellowship in Population Dynamics: Incorporating model selection and decision analysis into population dynamics modeling.* Grant awarded for \$120K.

Post-docs

2026- Zoe Rand (UW) Co-supervised with Stephanie Moore, Ole Shelton, Ryan Kelly

2023-2024 Megan Feddern (UAF) Co-supervised with Curry Cunningham, Mary Hunsicker, Will Satterthwaite

2021-2023 Alex Jensen (UW) Co-supervised with Ryan Kelly, Ole Shelton, Will Satterthwaite

2019-2021 Christian Commander (NWFSC) Co-supervised with Lewis Barnett and Tim Essington

2017-2021 Ben Nelson (NWFSC) Co-supervised with Ole Shelton

2017-2019 Lewis Barnett (NWFSC)

2015-2017 Sean Anderson (Smith Fellow). Co-supervised with Trevor Branch

2012-2013 Guillaume Bal (NRC Research Post-doc). Co-supervised with Eli Holmes and Mark Scheuerell

2012 Jim Thorson (NRC Research Post-doc). With Michelle McClure, Eli Holmes, and Mark Scheuerell

Supervised Graduate Students or Committees

2021–2025 Sierra Gilman (B. Gardner), SEFS, University of Washington

2019–2022 Sam May (K. Naish), SAFS, University of Washington

2017–2021 Natalie Lowell (L. Hauser), SAFS, University of Washington (NMFS Population Dynamics Fellowship)

2017–2021 Megan Feddern (G. Holtgrieve), SAFS, University of Washington

2017–2020 Elizabeth Ng (T. Essington), SAFS, University of Washington (NMFS Population Dynamics Fellowship)

2012–2018 Pam Moriarty (T. Essington), SAFS, University of Washington

2013–2016 Brandon Chasco (S. Heppell) Oregon State University (NMFS Population Dynamics Fellowship)

2014–2015 Brian Stock (B. Semmens) SIO, with Tomo Eguchi

2015–2016 Hannah Linder (J. Horne) UW

Service and Outreach

Subject Editor, Ecological Applications 2013 - 2025

Subject Editor, PeerJ 2020 - present

Subject Editor, Methods in Ecology and Evolution 2022 - present

DFO - NOAA climate change working group 2021 - present

NOAA Fisheries Integrated Toolbox co-lead 2021 - present

Puget Sound Seabird Survey 2005-2021
Developed survey methodology for Puget Sound's first land-based citizen science seabird survey and oversaw data analysis

Orca Bowl Washington State Sea Grant / University of Washington

Secretary & Treasurer, Statistical Ecology Section, Ecological Society of America 2010-2012